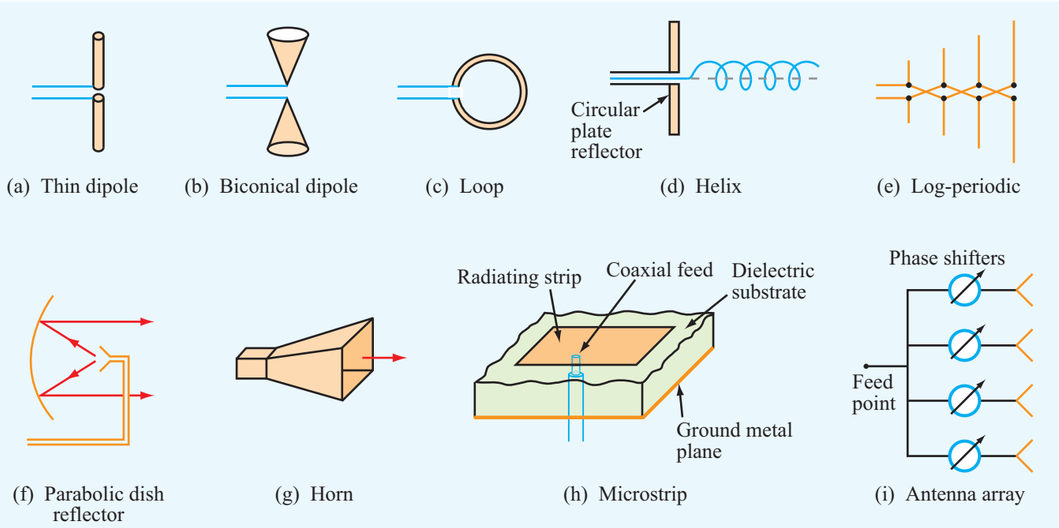


Figure 9-2 Various types of antennas.

Radiation Sources

Radiation sources fall into two categories: currents and aperture fields. The dipole and loop antennas (**Fig. 9-2(a)** and **(c)**) are examples of current sources; the time-varying currents flowing in the conducting wires give rise to the radiated electromagnetic fields. A horn antenna (**Fig. 9-2(g)**) is an example of the second group because the electric and magnetic fields across the horn's aperture serve as the sources of the radiated fields. The aperture fields are themselves induced by time-varying currents on the surfaces of the horn's walls. Therefore, all radiation ultimately is due to time-varying currents. The choice of currents or apertures as the sources is merely a computational convenience arising from the structure of the antenna. We will examine the radiation processes associated with both types of sources.

Far-Field Region

The wave radiated by a point source is spherical in nature, with the wavefront expanding outward at a rate equal to the phase velocity u_p (or the velocity of light c if the medium is free space). If R, which is the distance between the transmitting antenna and the receiving antenna, is sufficiently large enough for the wavefront across the receiving aperture to be considered planar (**Fig. 9-3**), then the receiving aperture is said to be

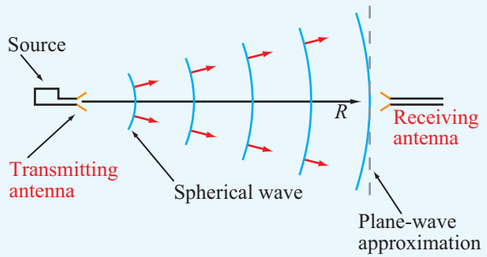


Figure 9-3 Far-field plane-wave approximation.

in the ***far-field*** (or ***far-zone***) region of the transmitting point source. This region is of particular significance because—for most applications—the location of the observation point is indeed in the far-field region of the antenna. The far-field plane-wave approximation allows the use of certain mathematical approximations that simplify the computation of the radiated field and, conversely, provide convenient techniques for synthesizing the appropriate antenna structure that would give rise to the desired far-field antenna pattern.

386 CHAPTER 9 RADIATION AND ANTENNAS

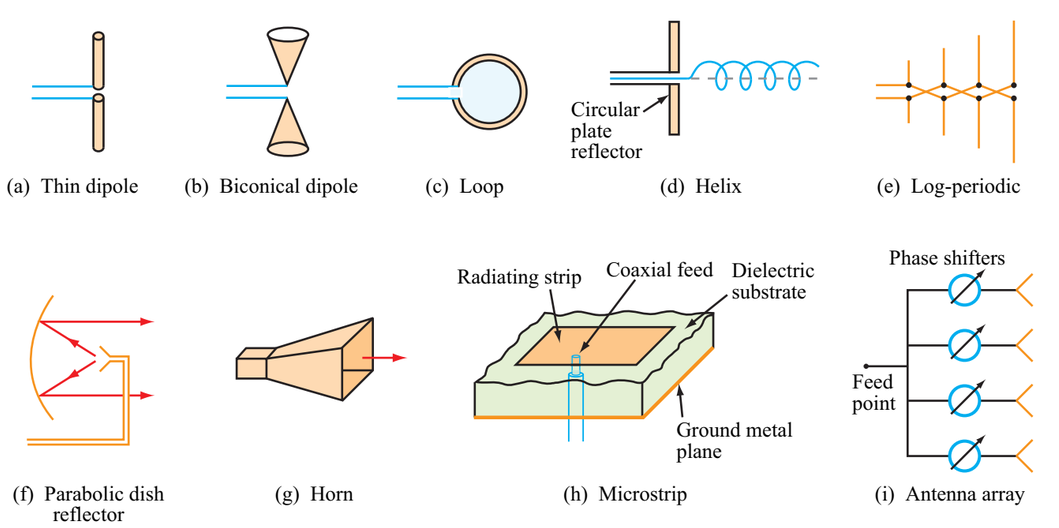


图9-2 各种类型的天线。

辐射源

辐射源分为两类：电流和孔径场。偶极子天线和环形天线（图9-2（a）和（c））是电流源的示例；导线中流动的时变电流会产生辐射电磁场。喇叭天线（图 9-2(g)）是第二组的一个例子，因为喇叭孔径上的电场和磁场充当辐射场的来源。孔径场本身是由喇叭壁表面上的时变电流引起的。因此，所有辐射最终都是由于时变电流造成的。选择电流或孔径作为源只是出于天线结构的计算方便。我们将检查与这两种类型的源相关的辐射过程。

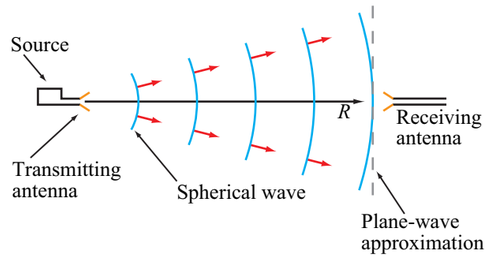


图 9-3 远场平面波近似。

远场区

点源辐射的波本质上是球形的，波前以等于相速度 u_p（或光速 c（如果介质是自由空间））的速率向外扩展。如果 R（发射天线和接收天线之间的距离）足够大，足以使穿过接收孔径的波前被视为平面（图 9-3），则接收孔径称为

在发射点源的远场（或远区）区域。该区域具有特别重要的意义，因为对于大多数应用而言，观察点的位置确实位于天线的远场区域。远场平面波近似允许使用某些数学近似来简化辐射场的计算，相反，提供了合成适当天线结构的便捷技术，从而产生所需的远场天线方向图。

Antenna Arrays

When multiple antennas operate together, the combination is called an ***antenna array*** (**Fig. 9-2(i)**), and the array as a whole behaves as if it were a single antenna. By controlling the magnitude and phase of the signal feeding each antenna, it is possible to *shape the radiation pattern* of the array and to *electronically steer the direction of the beam electronically.* These topics are treated in Sections 9-9 to 9-11.

9-1 The Hertzian Dipole

By regarding a linear antenna as consisting of a large number of infinitesimally short conducting elements where each is so short that current may be considered uniform over its length, the field of the entire antenna may be obtained by integrating the fields from all these differential antennas with the proper magnitudes and phases taken into account. We shall first examine the radiation properties of such a differential antenna, known as a ***Hertzian dipole***, and in Section 9-3, we will extend the results to compute the fields radiated by a half-wave dipole, which is commonly used as a standard antenna for many applications.

► A ***Hertzian dipole*** is a thin, linear conductor whose length l is very short compared with the wavelength λ; l should not exceed $\lambda/50$. ◄

In this case, the wire is oriented along the z direction in **Fig. 9-4** and carries a sinusoidally varying current given by

$$i(t) = I_0 \cos \omega t = \Re\mathfrak{e}[I_0 e^{j\omega t}] \quad \text{(A)}, \tag{9.1}$$

where I_0 is the current amplitude. From Eq. (9.1), the phasor current $\tilde{I} = I_0$. Even though the current has to go to zero at the two ends of the dipole, we shall treat it as constant across its entire length.

The customary approach for finding the electric and magnetic fields at a point Q in space (**Fig. 9-4**) due to radiation by a current source is through the retarded vector potential $\mathbf{A}$. From Eq. (6.84), the phasor retarded vector potential $\tilde{\mathbf{A}}(R)$ at a distance vector $\mathbf{R}$ from a volume υ' containing a phasor current distribution $\tilde{\mathbf{J}}$ is given by

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu_0}{4\pi} \int_{\upsilon'} \frac{\tilde{\mathbf{J}} e^{-jkR'}}{R'} \, d\upsilon', \tag{9.2}$$

where μ_0 is the magnetic permeability of free space (because the observation point is in air) and $k = \omega/c = 2\pi/\lambda$ is the wavenumber. For the dipole, the current density is simply

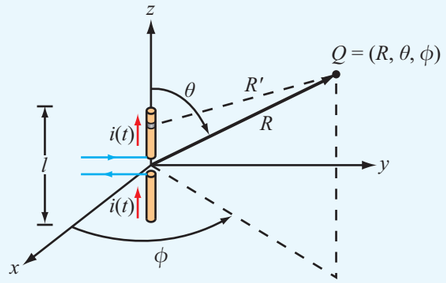


Figure 9-4 Short dipole placed at the origin of a spherical coordinate system.

$\tilde{\mathbf{J}} = \hat{\mathbf{z}}(I_0/s)$, where s is the cross-sectional area of the dipole wire. Also, $d\upsilon' = s\,dz$, and the limits of integration are from $z = -l/2$ to $z = l/2$. In **Fig. 9-4**, the distance R' between the observation point and a given point along the dipole is not the same as the distance to its center, R, but because we are dealing with a very short dipole, we can set $R' \approx R$. Hence,

$$\tilde{\mathbf{A}} = \frac{\mu_0}{4\pi} \frac{e^{-jkR}}{R} \int_{-l/2}^{l/2} \hat{\mathbf{z}} I_0 \, dz = \hat{\mathbf{z}} \frac{\mu_0}{4\pi} I_0 l \left(\frac{e^{-jkR}}{R} \right), \tag{9.3}$$

► The function (e^{-jkR}/R) is called the ***spherical propagation factor***. It accounts for the $1/R$ decay of the magnitude with distance as well as the phase change represented by e^{-jkR}. ◄

The direction of $\tilde{\mathbf{A}}$ is the same as that of the current (z direction).

Because our objective is to characterize the directional character of the radiated power at a fixed distance R from the antenna, antenna pattern plots are presented in a spherical coordinate system (**Fig. 9-5**). Its variables, R, θ, and ϕ, are called the ***range***, ***zenith angle***, and ***azimuth angle***, respectively. To that end, we need to write $\tilde{\mathbf{A}}$ in terms of its spherical coordinate components, which is realized with the help of Eq. (3.65c) by expressing $\hat{\mathbf{z}}$ in terms of spherical coordinates:

$$\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos\theta - \hat{\boldsymbol{\theta}} \sin\theta. \tag{9.4}$$

Upon substituting Eq. (9.4) into Eq. (9.3), we obtain

$$\tilde{\mathbf{A}} = (\hat{\mathbf{R}} \cos\theta - \hat{\boldsymbol{\theta}} \sin\theta) \frac{\mu_0 I_0 l}{4\pi} \left(\frac{e^{-jkR}}{R} \right) = \hat{\mathbf{R}} \tilde{A}_R + \hat{\boldsymbol{\theta}} \tilde{A}_\theta + \hat{\boldsymbol{\phi}} \tilde{A}_\phi, \tag{9.5}$$

天线阵列

当多个天线一起工作时，这种组合称为天线阵列（图9-2（i）），整个阵列的行为就像单个天线一样。通过控制馈送到每个天线的信号的幅度和相位，可以塑造阵列的辐射方向图并以电子方式控制波束的方向。这些主题将在第 9-9 至 9-11 节中讨论。

9-1 赫兹偶极子

通过将线性天线视为由大量无限短的导电元件组成，其中每个元件都很短，以至于可以认为电流在其长度上是均匀的，则可以通过对所有这些差分天线的场进行积分并考虑适当的幅度和相位来获得整个天线的场。我们将首先检查这种差分天线（称为赫兹偶极子）的辐射特性，在第 9-3 节中，我们将扩展结果以计算半波偶极子辐射的场，半波偶极子通常用作许多应用的标准天线。

► 赫兹偶极子是一种薄的线性导体，其长度 l 与波长 λ 相比非常短； l 不应超过λ/50。 ◄

在这种情况下，导线沿图 9-4 中的 z 方向定向，并承载由下式给出的正弦变化电流：

$$i(t) = I_0 \cos \omega t = \Re\mathfrak{e}[I_0 e^{j\omega t}] \quad \text{(A)}, \tag{9.1}$$

其中 I_0 是电流幅度。从方程。 (9.1) 相量电流 $\tilde{I} = I_0$。尽管偶极子两端的电流必须为零，但我们仍将其视为整个长度上的常数。

寻找空间 Q 点（图 9-4）由于电流源辐射而产生的电场和磁场的常规方法是通过延迟矢量势 A。 (6.84)，距包含相量电流分布 $\tilde{J}$ 的体积 υ' 距离矢量 R 处的相量延迟矢量势 $\tilde{A}$(R) 由下式给出

$$\tilde{\mathbf{A}}(\mathbf{R}) = \frac{\mu_0}{4\pi} \int_{\upsilon'} \frac{\tilde{\mathbf{J}} e^{-jkR'}}{R'} \, d\upsilon', \tag{9.2}$$

其中 μ_0 是自由空间的磁导率（因为观察点在空气中），$k = \omega/c = 2\pi/\lambda$ 是波数。对于偶极子，电流密度很简单

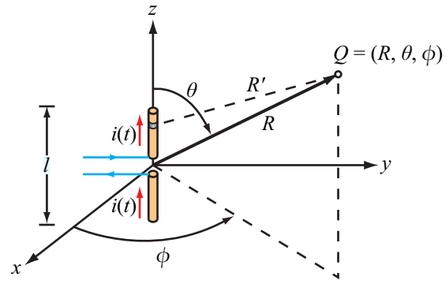


图 9-4 放置在球坐标系原点的短偶极子。

$\tilde{J} = \hat{z}(I_0/s)$，其中 s 是偶极线的横截面积。另外，$d\upsilon' = s\,dz$，积分的极限是从 $z = -l/2$ 到 $z = l/2$。在图 9-4 中，观察点与偶极子上给定点之间的距离 R′ 与到其中心 R 的距离不同，但因为我们处理的是非常短的偶极子，所以可以设置 $R' \approx R$。因此，

$$\tilde{\mathbf{A}} = \frac{\mu_0}{4\pi} \frac{e^{-jkR}}{R} \int_{-l/2}^{l/2} \hat{\mathbf{z}} I_0 \, dz = \hat{\mathbf{z}} \frac{\mu_0}{4\pi} I_0 l \left(\frac{e^{-jkR}}{R} \right), \tag{9.3}$$

► 函数 (e^{-jkR}/R) 称为球面传播因子。它解释了 $1/R$ 幅度随距离的衰减以及由 e^{-jkR} 表示的相位变化。 ◄

$\tilde{A}$ 的方向与电流方向（z 方向）相同。

因为我们的目标是表征距天线固定距离 R 处的辐射功率的方向特性，所以天线方向图以球坐标系呈现（图 9-5）。它的变量 R、θ 和 φ 分别称为距离、天顶角和方位角。为此，我们需要将 $\tilde{A}$ 写成其球坐标分量，这可以借助方程 1 来实现。 (3.65c) 用球坐标表示 $\hat{z}$：

$$\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos\theta - \hat{\boldsymbol{\theta}} \sin\theta. \tag{9.4}$$

代入方程式后。 (9.4) 代入等式。 (9.3)，我们得到

$$\tilde{\mathbf{A}} = (\hat{\mathbf{R}} \cos\theta - \hat{\boldsymbol{\theta}} \sin\theta) \frac{\mu_0 I_0 l}{4\pi} \left(\frac{e^{-jkR}}{R} \right) = \hat{\mathbf{R}} \tilde{A}_R + \hat{\boldsymbol{\theta}} \tilde{A}_\theta + \hat{\boldsymbol{\phi}} \tilde{A}_\phi, \tag{9.5}$$

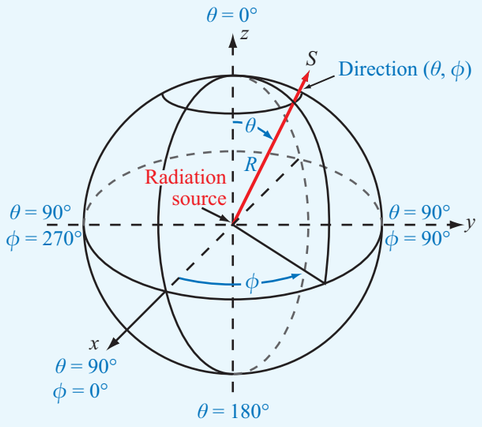


Figure 9-5 Spherical coordinate system.

with

$$\widetilde{A}_R = \frac{\mu_0 I_0 l}{4\pi}\cos\theta\left(\frac{e^{-jkR}}{R}\right), \tag{9.6a}$$

$$\widetilde{A}_\theta = -\frac{\mu_0 I_0 l}{4\pi}\sin\theta\left(\frac{e^{-jkR}}{R}\right), \tag{9.6b}$$

$$\widetilde{A}_\phi = 0.$$

With the spherical components of $\widetilde{\mathbf{A}}$ known, the next step is straightforward; we simply apply the free-space relationships given by Eqs. (6.85) and (6.86) as

$$\widetilde{\mathbf{H}} = \frac{1}{\mu_0}\nabla\times\widetilde{\mathbf{A}} \tag{9.7a}$$

and

$$\widetilde{\mathbf{E}} = \frac{1}{j\omega\varepsilon_0}\nabla\times\widetilde{\mathbf{H}} \tag{9.7b}$$

to obtain the expressions

$$\widetilde{H}_\phi = \frac{I_0 lk^2}{4\pi}e^{-jkR}\left[\frac{j}{kR}+\frac{1}{(kR)^2}\right]\sin\theta, \tag{9.8a}$$

$$\widetilde{E}_R = \frac{2I_0 lk^2}{4\pi}\eta_0 e^{-jkR}\left[\frac{1}{(kR)^2}-\frac{j}{(kR)^3}\right]\cos\theta, \tag{9.8b}$$

$$\widetilde{E}_\theta = \frac{I_0 lk^2}{4\pi}\eta_0 e^{-jkR}\left[\frac{j}{kR}+\frac{1}{(kR)^2}-\frac{j}{(kR)^3}\right]\sin\theta, \tag{9.8c}$$

where $\eta_0 = \sqrt{\mu_0/\varepsilon_0} \approx 120\pi\ (\Omega)$ is the intrinsic impedance of free space. The remaining components ($\widetilde{H}_R$, $\widetilde{H}_\theta$, and $\widetilde{E}_\phi$) are everywhere zero. **Figure 9-6** depicts the electric field lines of the wave radiated by the short dipole.

9-1.1 Far-Field Approximation

As was stated earlier, in most antenna applications, we are primarily interested in the radiation pattern of the antenna at great distances from the source. For the electric dipole, this corresponds to distances R so that $R \gg \lambda$ or, equivalently, $kR = 2\pi R/\lambda \gg 1$. This condition allows us to neglect the terms varying as $1/(kR)^2$ and $1/(kR)^3$ in Eqs. (9.8a) to (9.8c) in favor of the terms varying as $1/kR$, which yields the far-field expressions

$$\widetilde{E}_\theta = \frac{jI_0 lk\eta_0}{4\pi}\left(\frac{e^{-jkR}}{R}\right)\sin\theta \quad (\text{V/m}), \tag{9.9a}$$

$$\widetilde{H}_\phi = \frac{\widetilde{E}_\theta}{\eta_0} \quad (\text{A/m}), \tag{9.9b}$$

and $\widetilde{E}_R$ is negligible. At the observation point Q (**Fig. 9-4**), the wave now appears similar to a uniform plane wave with its electric and magnetic fields in phase, related by the intrinsic impedance of the medium η_0, and their directions orthogonal to each other and to the direction of propagation ($\hat{\mathbf{R}}$). Both fields are proportional to $\sin\theta$ and independent of ϕ (which is expected from symmetry considerations).

9-1.2 Power Density

Given $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$, the ***time-average Poynting vector*** of the radiated wave, which is also called the ***power density***, can be obtained by applying Eq. (7.100); that is,

$$\mathbf{S}_{\text{av}} = \tfrac{1}{2}\mathfrak{Re}\left(\widetilde{\mathbf{E}}\times\widetilde{\mathbf{H}}^*\right) \quad (\text{W/m}^2). \tag{9.10}$$

For the short dipole, use of Eqs. (9.9a) and (9.9b) yields

$$\mathbf{S}_{\text{av}} = \hat{\mathbf{R}}\,S(R,\theta), \tag{9.11}$$

388 CHAPTER 9 RADIATION AND ANTENNAS

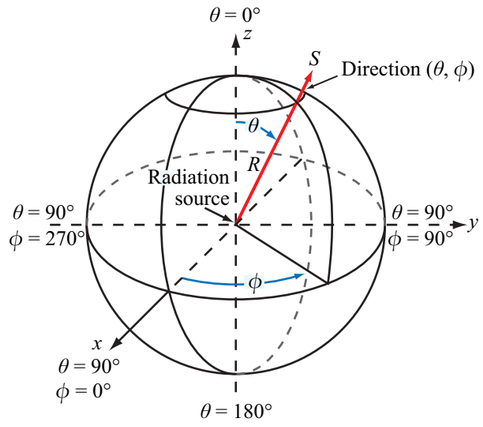


图 9-5 球坐标系。

其中 $\eta_0 = \sqrt{\mu_0/\varepsilon_0} \approx 120\pi\ (\Omega)$ 是自由空间的固有阻抗。其余分量（$\widetilde{H}_R$、$\widetilde{H}_\theta$ 和 $\widetilde{E}_\phi$）处处为零。图 9-6 描绘了短偶极子辐射的电场线。

9-1.1 远场近似

如前所述，在大多数天线应用中，我们主要对距源较远的天线的辐射方向图感兴趣。对于电偶极子，这对应于距离 R，使得 $R \gg \lambda$ 或等效地 $kR = 2\pi R/\lambda \gg 1$。这个条件允许我们忽略等式中 $1/(kR)^2$ 和 $1/(kR)^3$ 变化的项。(9.8a) 到 (9.8c) 支持项变化为 $1/kR$，从而产生远场表达式

$$\widetilde{E}_\theta = \frac{jI_0 lk\eta_0}{4\pi}\left(\frac{e^{-jkR}}{R}\right)\sin\theta \quad (\text{V/m}), \tag{9.9a}$$

$$\widetilde{H}_\phi = \frac{\widetilde{E}_\theta}{\eta_0} \quad (\text{A/m}), \tag{9.9b}$$

并且 $\widetilde{E}_R$ 可以忽略不计。在观察点 Q（图 9-4），波现在看起来类似于均匀平面波，其电场和磁场同相，与介质的固有阻抗 η_0 相关，并且它们的方向彼此正交并与传播方向 ($\hat{R}$) 正交。两个场都与 $\sin\theta$ 成正比，并且与 ϕ（无关，这是出于对称性考虑而预期的）。

已知 $\widetilde{A}$ 的球形分量后，下一步就很简单了；我们简单地应用方程给出的自由空间关系。(6.85) 和 (6.86) 为

$$\widetilde{A}_R = \frac{\mu_0 I_0 l}{4\pi}\cos\theta\left(\frac{e^{-jkR}}{R}\right), \tag{9.6a}$$

$$\widetilde{A}_\theta = -\frac{\mu_0 I_0 l}{4\pi}\sin\theta\left(\frac{e^{-jkR}}{R}\right), \tag{9.6b}$$

$$\widetilde{A}_\phi = 0.$$

$$\widetilde{\mathbf{H}} = \frac{1}{\mu_0}\nabla\times\widetilde{\mathbf{A}} \tag{9.7a}$$

和

$$\widetilde{\mathbf{E}} = \frac{1}{j\omega\varepsilon_0}\nabla\times\widetilde{\mathbf{H}} \tag{9.7b}$$

获得表达式

$$\widetilde{H}_\phi = \frac{I_0 lk^2}{4\pi}e^{-jkR}\left[\frac{j}{kR}+\frac{1}{(kR)^2}\right]\sin\theta, \tag{9.8a}$$

$$\widetilde{E}_R = \frac{2I_0 lk^2}{4\pi}\eta_0 e^{-jkR}\left[\frac{1}{(kR)^2}-\frac{j}{(kR)^3}\right]\cos\theta, \tag{9.8b}$$

$$\widetilde{E}_\theta = \frac{I_0 lk^2}{4\pi}\eta_0 e^{-jkR}\left[\frac{j}{kR}+\frac{1}{(kR)^2}-\frac{j}{(kR)^3}\right]\sin\theta, \tag{9.8c}$$

9-1.2 功率密度

给定 $\widetilde{E}$ 和 $\widetilde{H}$，辐射波的时均坡印廷矢量，也称为功率密度，可以通过应用式（1）获得。(7.100)；那是，

$$\mathbf{S}_{\text{av}} = \tfrac{1}{2}\mathfrak{Re}\left(\widetilde{\mathbf{E}}\times\widetilde{\mathbf{H}}^*\right) \quad (\text{W/m}^2). \tag{9.10}$$

对于短偶极子，使用方程。(9.9a) 和 (9.9b) 产量

$$\mathbf{S}_{\text{av}} = \hat{\mathbf{R}}\,S(R,\theta), \tag{9.11}$$

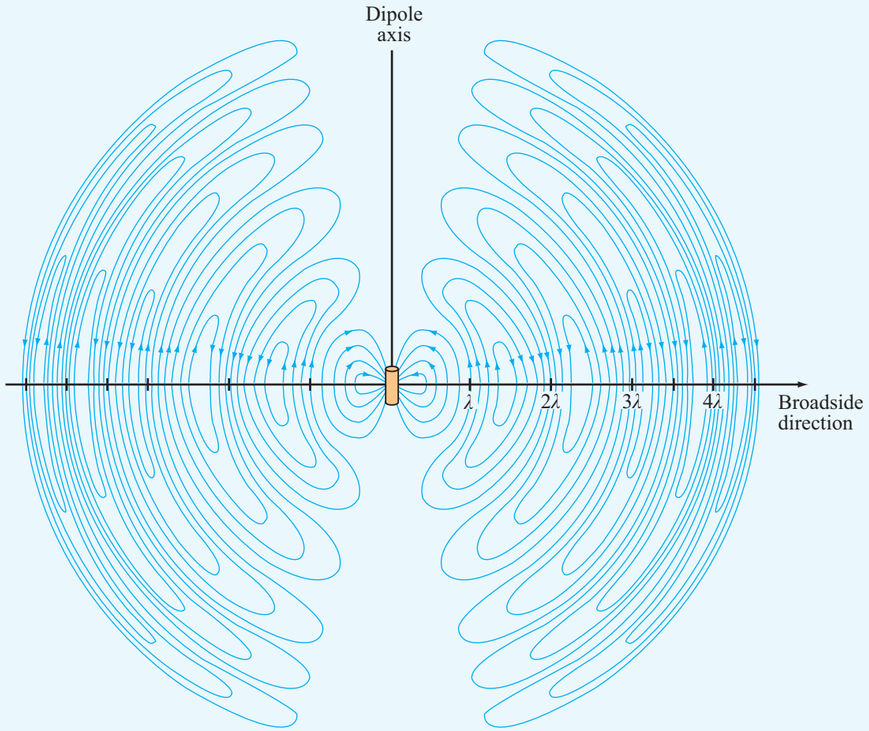


Figure 9-6 Electric field lines surrounding an oscillating dipole at a given instant.

with

$$S(R,\theta) = \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\right)\sin^2\theta = S_0 \sin^2\theta \qquad (\mathrm{W/m^2}). \tag{9.12}$$

The directional pattern of any antenna is described in terms of the ***normalized radiation intensity*** $F(\theta,\phi)$ and is defined as the ratio of the power density $S(R,\theta,\phi)$ at a specified range R to $S_{\max}$, which is the maximum value of $S(R,\theta,\phi)$ at the same range. Thus,

$$F(\theta,\phi) = \frac{S(R,\theta,\phi)}{S_{\max}}. \qquad \text{(dimensionless)} \tag{9.13}$$

For the Hertzian dipole, the $\sin^2\theta$ dependence in Eq. (9.12) indicates that the radiation is maximum in the broadside direction ($\theta = \pi/2$), corresponding to the azimuth plane, and is given by

$$S_{\max} = S_0 = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} = \frac{15\pi I_0^2}{R^2}\left(\frac{l}{\lambda}\right)^2 \qquad (\mathrm{W/m^2}), \tag{9.14}$$

where use was made of the relations $k = 2\pi/\lambda$ and $\eta_0 \approx 120\pi$. We observe that $S_{\max}$ is directly proportional to I_0^2 and l^2 (with l measured in wavelengths) and that it decreases with distance as $1/R^2$.

From the definition of the normalized radiation intensity given by Eq. (9.13), it follows that

$$F(\theta,\phi) = F(\theta) = \sin^2\theta. \tag{9.15}$$

Plots of $F(\theta)$ are shown in **Fig. 9-7** in both the elevation plane (the θ plane) and the azimuth plane (ϕ plane).

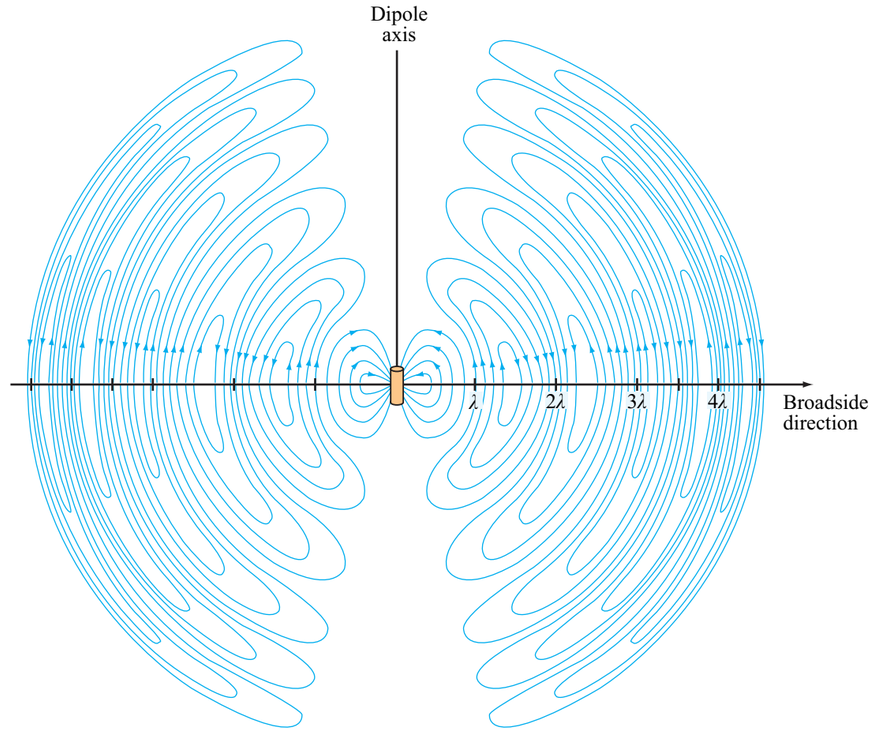


图 9-6 在给定时刻围绕振荡偶极子的电场线。

和

$$S(R,\theta) = \left(\frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\right)\sin^2\theta = S_0 \sin^2\theta \qquad (\mathrm{W/m^2}). \tag{9.12}$$

任何天线的方向图都用归一化辐射强度 $F(\theta,\phi)$ 来描述，并定义为在指定范围 R 到 S_{max} 上功率密度 $S(R,\theta,\phi)$ 的比率，它是同一范围内 $S(R,\theta,\phi)$ 的最大值。因此，

$$F(\theta,\phi) = \frac{S(R,\theta,\phi)}{S_{\max}}. \qquad \text{(dimensionless)} \tag{9.13}$$

对于赫兹偶极子，方程中的 $\sin^2\theta$ 依赖性： (9.12) 表示辐射在对应于方位平面的侧边方向 ($\theta = \pi/2$) 上最大，由下式给出

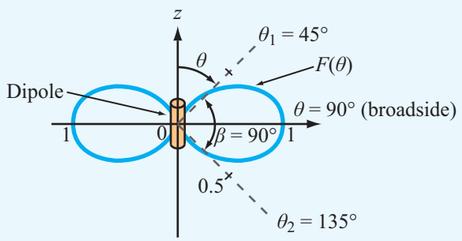
$$S_{\max} = S_0 = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} = \frac{15\pi I_0^2}{R^2}\left(\frac{l}{\lambda}\right)^2 \qquad (\mathrm{W/m^2}), \tag{9.14}$$

其中使用关系 $k = 2\pi/\lambda$ 和 $\eta_0 \approx 120\pi$。我们观察到 S_{max} 与 I_0^2 和 l^2（成正比，其中 l 以波长）测量，并且它随着距离的增加而减小，即 $1/R^2$。

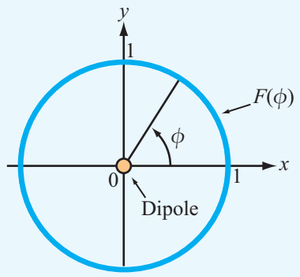
根据方程给出的归一化辐射强度的定义： (9.13)，由此可知

$$F(\theta,\phi) = F(\theta) = \sin^2\theta. \tag{9.15}$$

$F(\theta)$ 的图如图 9-7 所示，在仰角平面（θ 平面）和方位角平面（φ 平面）中。



(a) Elevation pattern



(b) Azimuth pattern

Figure 9-7 Radiation patterns of a short dipole.

► No energy is radiated by the short dipole along the direction of the dipole axis, and maximum radiation ($F = 1$) occurs in the ***broadside direction*** ($\theta = 90°$). Since $F(\theta)$ is independent of ϕ, the pattern is doughnut-shaped in θ–ϕ space. ◄

Concept Question 9-1: What does it mean to say that most antennas are reciprocal devices?

Concept Question 9-2: What is the radiated wave like in the far-field region of the antenna?

Concept Question 9-3: In a Hertzian dipole, what is the underlying assumption about the current flowing through the wire?

Concept Question 9-4: Outline the basic steps used to relate the current in a wire to the radiated power density.

Exercise 9-1: A 1-m long dipole is excited by a 5 MHz current with an amplitude of 5 A. At a distance of 2 km, what is the power density radiated by the antenna along its broadside direction?

Answer: $S_0 = 8.2 \times 10^{-8}$ W/m^2. (See EM.)

9-2 Antenna Radiation Characteristics

An ***antenna pattern*** describes the far-field directional properties of an antenna when measured at a fixed distance from the antenna. In general, the antenna pattern is a three-dimensional plot that displays the strength of the radiated field or power density as a function of direction—with direction being specified by the zenith angle θ and the azimuth angle ϕ.

► By virtue of reciprocity, a receiving antenna has the same directional antenna pattern as the pattern that it exhibits when operated in the transmission mode. ◄

Consider a transmitting antenna placed at the origin of the observation sphere shown in **Fig. 9-8**. The differential power radiated by the antenna through an elemental area dA is

$$dP_{\text{rad}} = \mathbf{S}_{\text{av}} \cdot d\mathbf{A} = \mathbf{S}_{\text{av}} \cdot \hat{\mathbf{R}}\, dA = S\, dA \qquad \text{(W)}, \qquad (9.16)$$

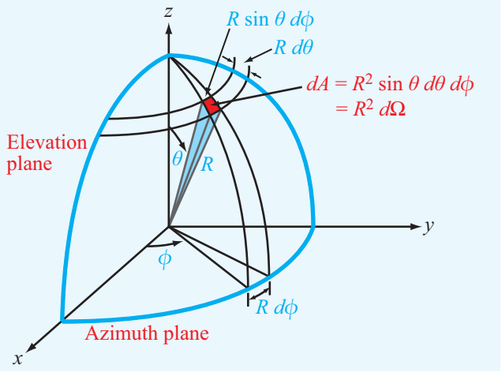
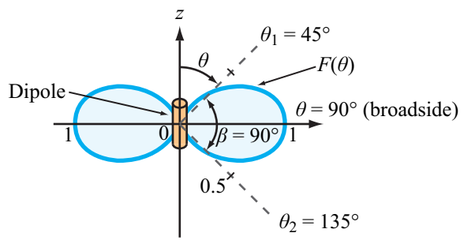
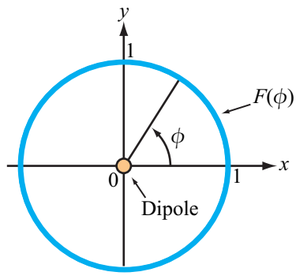


Figure 9-8 Definition of solid angle $d\Omega = \sin\theta\, d\theta\, d\phi$.



(a) Elevation pattern



(b) Azimuth pattern

图 9-7 短偶极子的辐射方向图。

► 短偶极子沿偶极轴方向不辐射能量，最大辐射（F = 1）出现在侧边方向（θ = 90°）。由于 F(θ) 独立于 ϕ，因此该模式在 θ–ϕ 空间中呈环形。◄

概念问题 9-1：大多数天线是互易设备是什么意思？

概念问题 9-2：天线远场区域的辐射波是什么样的？

概念问题 9-3：在赫兹偶极子中，关于流经电线的电流的基本假设是什么？

概念问题 9-4：概述用于将电线中的电流与辐射功率密度联系起来的基本步骤。

练习9-1：一个1米长的偶极子被5MHz、5A幅值的电流激励。在2km距离处，天线沿其侧面方向辐射的功率密度是多少？

答案：$S_0 = 8.2 \times 10^{-8}$ W/m^2。（参见EM。）

9-2 天线辐射特性

天线方向图描述了在距天线固定距离处测量时天线的远场方向特性。一般来说，天线方向图是一个三维图，显示辐射场的强度或功率密度作为方向的函数，方向由天顶角 θ 和方位角 ϕ 指定。

► 由于互易性，接收天线具有与其在传输模式下操作时表现出的方　向图相同的方向性天线方向图。◄

考虑如图 9-8 所示放置在观测球原点的发射天线。天线通过单元区域辐射的差分功率 dA 为

$$dP_{\text{rad}} = \mathbf{S}_{\text{av}} \cdot d\mathbf{A} = \mathbf{S}_{\text{av}} \cdot \hat{\mathbf{R}}\, dA = S\, dA \qquad \text{(W)}, \qquad (9.16)$$

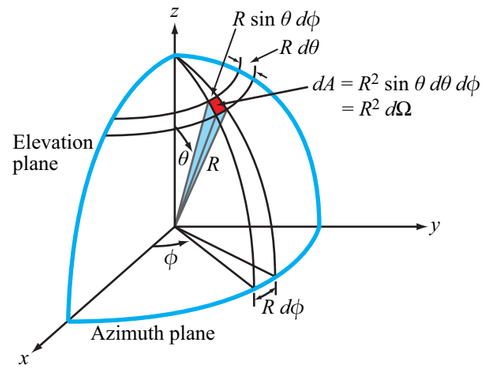


图 9-8 立体角 dΩ = sin θ dθ dϕ 的定义。